

Why Measure Theory

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1 Introduction: Motivation

1.1 Definitions

I will begin with the definition of Riemann Integration.

Definition 1 (Reimann Integration). Let f be a bounded function $f: [a, b] \rightarrow \mathbb{R}$. Let P be a partition of the domain of the form

$$P = \{x_0, x_1, \dots, x_n\}$$

Such that

$$a = x_0 < x_1 < \dots < x_n = b$$

If for every $\epsilon > 0$ there exists a partition P_ϵ such that $U(f, P_\epsilon) - L(f, P_\epsilon) < \epsilon$ then $f(x)$ is Riemann Integrable with $\int f = U(f) = L(f)$.

Where $U(f, P)$ is the upper sums

$$U(f, P_\epsilon) = \sum_{k=1}^n (x_k - x_{k-1}) \sup_{[x_{k-1}, x_k]} f$$

and $U(f)$ is inf over all partitions P

$$U(f) = \inf_P U(f, P)$$

Where $L(f, P)$ is the lower sums

$$L(f, P) = \sum_{k=1}^n (x_k - x_{k-1}) \inf_{[x_{k-1}, x_k]} f$$

and $L(f)$ is sup over all partitions P

$$L(f) = \sup_P L(f, P)$$

(Show that for a function on $[0, 1]$ enumerating the rationals converging to dirchlet the function at the limit isnt in the class of riemann integrable functions we know this because lebesgue defines whats riemann integrable; to get there explain measure 0 first don't give the entire lebesgue measure definition in the motivation, say you can integrate it with respect to measure then later explain through the paper; and explain where that fails and later why sigma algebra and then vitalli and so on.

1.2 Riemann Failure

1.2.1 Function

Let the sequence of rational numbers $\{r_1, r_2, \dots\}$ enumerate the rationals on $[0, 1]$.

Let $f_n(x)$ be a series of functions of the form

$$f_n(x) = \begin{cases} 1 : & x \in \{r_1, r_2, \dots, r_n\} \\ 0 : & x \notin \{r_1, r_2, \dots, r_n\} \end{cases} \quad (1)$$

1.2.2 Upper and Lower Sums

Fix $n \in \mathbb{N}$.

Take the partition P_n such that $(x_k - x_{k-1}) \leq \frac{\epsilon}{4n}$ for all consecutive pairs of points in P_n .

The total number of intervals in which there are rational numbers is $2n$; since there are n rationals where each can be in at most 2 intervals at once (Footnote this maybe) (if $r_k = x_k$ for instance then $[x_{k-1}, x_k]$ and $[x_k, x_{k+1}]$ both contains r_k).

This gives

$$U(f_n, P_n) = \left(\frac{\epsilon}{4n}\right)(2n)(1) + (1)(0) = \frac{\epsilon}{2}$$

Where each length $(x_k - x_{k-1}) = \left(\frac{\epsilon}{4n}\right)$, there exists at most $2n$ of these intervals, each with $\sup_{[x_k - x_{k-1}]} f = 1$. All other possible intervals have a max combined length of 1, each with $\sup_{[x_k - x_{k-1}]} f = 0$.

Since $\mathbb{I}$ is dense in $\mathbb{R}$, between any two real numbers there exists an irrational. Meaning on any interval $[x_{k-1}, x_k]$, $\inf_{[x_{k-1}, x_k]} f = 0$ giving

$$L(f_n, P_n) = 0$$

This gives for any $\epsilon > 0$ for some $N \in \mathbb{N}$ whenever $n \geq N$ it follows

$$U(f_n, P_n) - L(f_n, P_n) = \frac{\epsilon}{2} < \epsilon$$

meaning the function is Riemann integrable with

$$\int_0^1 f_n = L(f_n) = 0$$

1.2.3 Limit

Since the sequence of values r_k are countable we have

$$\lim_{n \rightarrow \infty} f_n = \begin{cases} 1 : & x \in \mathbb{Q} \cap [0, 1] \\ 0 : & x \notin \mathbb{Q} \cap [0, 1] \end{cases} \quad (2)$$

Call this function $f(x)$

Now for any partition P since both $\mathbb{Q}$ is dense in $\mathbb{R}$ as well as $\mathbb{I}$ dense in $\mathbb{R}$ on any interval $[x_{k-1}, x_k] \in P$ there exists both a rational and irrational.

This gives $\inf_{[x_{k-1}, x_k]} f = 0$ as well $\sup_{[x_k - x_{k-1}]} f = 1$ on every interval. Which forces

$$U(f, P) = 1$$

and

$$L(f, P) = 0$$

for every partition P implying $U(f) = 1$ and $L(f) = 0$ which says that f is not Riemann integrable.

(Give measure 0 working here first and then explain how Riemann doesn't tell us of its own failure we need Lebesgue criterion, maybe start with the definition of measure 0 and briefly introduce outer measure)

1.3 Lebesgue Explanation

1.3.1 Measure 0

Definition 2 (Measure 0). A is a measure zero set if for every $\epsilon > 0$ there exists a countable covering $\bigcup_n O_n$ where each O_n is an open interval such that $A \subset \bigcup_n O_n$ where $\sum_n l(O_n) \leq \epsilon$. Where $l()$ denotes the length of a open interval defined as $b - a$ for any open interval (a, b) .

1.3.2 Lebesgue Criterion

By the Lebesgue criterion

Theorem 1 (Lebesgue Criterion). *A bounded function $f: [a, b] \rightarrow \mathbb{R}$ is Riemann integrable iff the set of discontinuities is of Measure 0*

1.3.3 f discontinuous everywhere

Recall that

$$\lim_{n \rightarrow \infty} f_n = \begin{cases} 1 & : x \in \mathbb{Q} \cap [0, 1] \\ 0 & : x \notin \mathbb{Q} \cap [0, 1] \end{cases}$$

Take any point $x \in [0, 1]$. Observe the definition of continuity.

Definition 3. A function f is continuous at a point c if for every $\epsilon > 0$ there exists a $\delta > 0$ such that whenever $|x - c| < \delta$ it follows that $|f(x) - f(c)| < \epsilon$

Now fix $\epsilon_0 \leq 1$ take $\epsilon_0 = \frac{1}{2}$. Choose any point $c \in [0, 1]$. Again both $\mathbb{Q}$ and $\mathbb{I}$ dense in $\mathbb{R}$.

First consider $c \in \mathbb{Q}$ then for any δ chosen there exists some irrational point $x \in (c - \delta, c + \delta)$ giving $|f(x) - f(c)| = 1 \geq \epsilon_0$.

Consider the case $c \in \mathbb{I}$ then for any δ chosen there exists some rational point $x \in (c - \delta, c + \delta)$ giving $|f(x) - f(c)| = 1 \geq \epsilon_0$.

Meaning the set of discontinuous points of f is all of $[0, 1]$ which has total measure 1.

Which explains why f is not Riemann integrable at the limit but Riemann integrable for every f_n .

1.4 Lebesgue Savior